

Economics 659 Behavioral Game Theory

Dr. John Van Huyck

The problem of strategic interdependence.

- A situation exhibits *strategic interdependence* when a decision maker's best strategy depends on the strategy adopted by related decision makers.
 - Game theory analyzes situations of strategic interdependence.
 - The self-referential nature of strategic interdependence makes it much more difficult to analyze than single agent decision problems.
 - Given strategic interdependence, optimality is not well defined in general.

Examples of Strategic Interdependence

- Rockets to the moon versus rockets to intercept a bogey.
- Airline workers hustling to get a plane away on time.
- Entrepreneurs deciding to enter a market.
- A baker discounting prices on pastries just before closing.
- People bidding for things on eBay.
- A government deciding to print money to pay off its creditors.
- Tennis players deciding to serve to the forehand or backhand.
- Other examples?

The Power of Game Theory

- Generality
 - All of the different situations of strategic interdependence are analyzed using the same basic ideas.
 - It can be used by all of the social sciences.
- Precise predictions.
 - It predicts exactly how frequently a server should serve to the forehand side. Field studies find professional tennis players conform to those frequencies.
 - It predicts exactly how many Entrepreneurs should enter a market. Predictions are consistent with observed behavior in the laboratory.

The Mutual Consistency Condition

- An *equilibrium* in a 'game' is defined as an 'assignment' to each player of a 'strategy' that is 'optimal' for him when the others use the strategies assigned to them.
- Key Words: game, assignment, strategy, and optimal.
 - What is a well-defined game?
 - When do equilibrium assignments exist?
 - Where do equilibrium assignments come from?
 - What is a strategy?
 - In what sense is a strategy optimal?

Also Provides a Prescriptive Theory

- Any prescriptive theory that effectively advises players how to play games must pick a mutually consistent strategy assignment in each game, that is, it must pick an equilibrium.
 - A theory recommending anything other than an equilibrium would be self-defeating in the sense that a player who believes that the others are following the theory will sometimes be motivated to deviate from it.
 - That is, the prescribed assignment would not be self-enforcing.

Example: Rousseau's Stag Hunt

- “If a group of hunters set out to take a stag, they are fully aware that they would all have to remain faithfully at their posts in order to succeed; but if a hare happens to pass near one of them, there can be no doubt that he pursued it without qualm, and that once he had caught his prey, he cared very little whether or not he had made his companions miss theirs.”
 - Quoted in Fudenberg/Tirole (1991, 3)

Strategic Form Representation of Stag Hunt

- A strategic form representation of a game has three objects:
 - A set of players
 - Feasible strategies for the players
 - The expected utility from all feasible strategy combinations for the players.
- Stag Hunt Example
 - The players $i \in \{1, 2, \dots, n\}$
 - The pure strategy space $S_i = \{1, 2\}$ for each player i ,
 - Where 1 denotes hunt rabbit and 2 denotes hunt stag.
 - Think of the index numbers as the level of effort.
 - let $u_i(s) = \min(s) - c \cdot s_i$, $1 > c > 0$.

Stag Hunt Example

- What prediction should one make about the outcome of the stag hunt game?

		Min(s_j)	
		1	2
Player i	1	1-c	1-c
	2	1-2c	2(1-c)

- Mutual Consistency
 - Everyone hunting stag is mutually consistent.
 - However, everyone hunting rabbits is also mutually consistent

Comparative Statics of n

- We might expect the likelihood of observing the efficient outcome, everyone hunt stag, to depend on the number of hunters, n ,
- that is, a few hunters succeed in catching the stag,
- While many hunters would fail to catch the stag.

		Min(s_{-i})	
		1	2
Player i	1	1-c	1-c
	2	1-2c	2(1-c)

Comparative Statics of c

- The likelihood of observing the efficient outcome should depend on the relative cost of individual effort, c ,
- that is, as c goes to 0 it becomes likely that everyone hunts stag,
- but as c goes to 1 it becomes likely that everyone hunts rabbit

		Min(s_{-i})	
		1	2
Player i	1	1-c	1-c
	2	1-2c	2(1-c)

Behavioral Game Theory

- Strategic Rationality is not well defined in the stag hunt game.
- Moreover, a behavioral theory sensitive to the amount of rationality required and the incentive to optimize will more closely fit our intuition about what to expect in the stag hunt.
- “Behavioral Game Theory is about what players actually do.”
 - Initial Behavior
 - Adaptive Behavior
 - Equilibrium Behavior

Game theory as a tool of the imagination

- It provides an analytical framework to think about situations that have never been observed before.
- Example: Spectrum auctions around the world.
 - Milgrom and Wilson used toy models from game theory to inform their imagination and design ascending multiple-unit auctions that had never been used before.
 - No one has been able to actually solve the auctions they designed.
- It can change ones understanding of familiar situations
 - For example, once one has played with the idea of spontaneous order one is much less willing to attribute economic prosperity to the President's or the Federal Reserve Chairman's talents.
 - Focus instead on institutions, that is, the rules of the game.

Game theory as a tool of reason

- Game theory is a mathematical subject
 - Almost all of the early game theorists were mathematicians rather than economists.
 - Mathematics, at least since Hilbert, transforms questions of truth into questions of consistency.
 - The central results are a collection of valid deductions.

A Science of Strategic Behavior

- Game theory provides a mathematical foundation for a science of strategic behavior.
- A Science of Strategic Behavior would provide effective prescriptions, true descriptions, and accurate predictions in situations of strategic interdependence.
- Science
 - “The process of replacing unordered masses of brute fact with tidy statements of orderly relations from which these facts can be inferred,” Simon.
 - “Independent, critical testing is a hallmark of what we call science,” McMullin.

Prescriptive Theory

- An effective prescriptive theory would prescribe the best way to achieve some objective in an observable game.
 - Game theory was originally conceived as a way to make effective prescriptions
 - See Harsanyi (1977) for an example in which game theory is taken seriously as a prescriptive theory.
- The fact that rational game theory may not be accurate raises the possibility of giving prescriptive advice.
- “Just as evangelists preach because people routinely violate moral codes, the fact that players violate game theory provides a chance to give helpful advice.” Camerer (2003,7)

Descriptive Theory

- A true descriptive theory would describe how people decide on how to behave in an observable game.
- Few use game theory in this way, but see the way neuroscientists are using game theory, McCabe (2002).
- Simon (1956) criticizes rational choice theory and, hence, game theory for not providing a theory of procedural rationality, but see Rubinstein’s book on *Modeling Bounded Rationality*.

Predictive Theory

- An accurate predictive theory would predict how actors behave in an observable game.
- Fudenberg & Tirole (1991) advocate this point of view, but don’t really take it seriously.
- Camerer (2003) demonstrates that an accurate theory will include behavioral assumptions derived from evidence in addition to purely rational assumptions.
- “The experimental evidence suggests that none of the leading theoretical frameworks for analyzing games—traditional non-cooperative game theory, cooperative game theory, evolutionary game theory, and adaptive learning models—gives a fully reliable account of behavior by itself, but that most behavior can be understood in terms of a synthesis of ideas from those frameworks, combined with empirical knowledge in proportions that depend in predictable ways on the environment.” Crawford (1997, 208) quoted in Camerer (2003, 22).


